

A Deep Learning Based Method for Identifying Protein Carbamylation Sites

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Abstract—Carbamylation is a lysine based post-translational modification (PTM) that plays a critical role in various biological mechanisms, such as CO₂ sensing. It has long been understudied due to the difficulties in detecting carbamate sites in proteins. Its unstable nature has rendered conventional experimental approaches ineffective. While computational methods were regarded as alternative ways, the lack of training data has limited the development of tools from popular computational frameworks. Recent advancements in experimental methodologies have uncovered a number of solvent accessible carbamate sites. These new discoveries made the application of computational models possible. The present work presented a curated dataset composed of newly-identified carbamate sites. Features were extracted from residue 3-d microenvironments, along with two biologically important measurements, pK_a and solvent accessible surface area (SASA). A deep learning model—a neural network featured by a Self-Attention mechanism based architecture was then trained. Using an independent test set, the network demonstrated its performance characterized by a 78.8% Sensitivity, a 70.6% Specificity, and a 75.0% accuracy. To evaluate the prevalence of Carbamylation in human and cyanobacteria proteins, large-scale model inferences were carried out. Results revealed that nearly 30% of the lysine residues in human and cyanobacteria proteins were predicted to be modifiable.

Index Terms—bioinformatics, post-translational modification, carbamylation, deep learning



1 INTRODUCTION

PROTEINS are chains of amino acids that fold into a specific 3-d structure based on their sequences (Branden and Tooze 2012). Post-translational modifications (PTMs) are local modifications of protein structures after their biosynthesis, in which modifying groups were attached to amino acid side chains (Ramazi and Zahiri 2021). PTMs are important because by slightly altering the structure of a protein, they can modulate its physicochemical properties and functions (K.-Y. Huang et al. 2019). As one of the most biologically important PTMs (Jimenez-Morales et al. 2014), Carbamylation is a modification that involves the covalent binding of carbon dioxide (CO₂) to the lysine side chain (Blake and Cann 2022). It is recognized as the key to “regulating oxygen-binding in haemoglobin and the activation of the CO₂-fixing enzyme RuBisCO” (Blake and Cann 2022). The study of this PTM facilitates the understanding of the biochemical mechanisms by which proteins interact with CO₂ (King et al. 2022).

In order for biologists to properly study such a modification, as many carbamate sites as possible should be determined. However, only on a limited number of proteins has this PTM been experimentally identified (King et al. 2022). It is suggested that Carbamylation is by no means well-studied due to the “lability and ready reversibility” of the modification (Linthwaite and Cann 2021). The lack of experimental hits partly stems from the spontaneous decomposition of the carbamylated lysine residues which makes the identification difficult (King et al. 2022). As pointed out by Linthwaite and Cann 2021, traditional experimental methods for identifying PTMs are too aggressive to be able to catch the labile carbamates. However, in the Protein Data Bank (PDB), a number of protein structures that contain carbamylated lysines can be found. These modification sites

are mostly special cases that the vast majority of them are metal co-ordinated ones with adjacent metal ions in the structures. In other words, these lysines are the “buried” ones. Nevertheless, they are the only cases that can be relatively easily observed at atomic resolution.

Computational methods can potentially make an important contribution to the field (Jimenez-Morales et al. 2014). However, despite the advantages of developing models that are faster than experimental ways and more suitable for large scale applications, there still remains a paucity of available tools in the literature. The major reason for the lack of proposed models is simply the scarcity of data as popular frameworks in Machine Learning or Deep Learning require big data.

It is the pivotal study carried out by Linthwaite, Janus, et al. 2018 that first proposed an experimental methodology for identifying carbamates that are solvent accessible (*i.e.* not buried), before that, systematical determination was not possible. Since then, more data has been experimentally uncovered. These discoveries, combined with a number of newly-identified sites in the recent work of King et al. 2022 have formed the basis of the application of computational methods. In light of the importance of the determination of this PTM and the accumulation of new data, the present work attempts to enrich the identification tools in the literature by proposing a computational method that leverages the Deep Learning technology.

PTM site identification can be viewed as a Deep Learning binary classification task in which a computational model is trained using known carbamate sites (Ground Truth) to classify a given lysine residue into two classes, *i.e.* modifiable or not modifiable. It is important to note that it is the 3-d structures of proteins that determines their biolog-

ical functions (Senior et al. 2020). Therefore, a protein 3-d structure (see Figure 1), specifically the microenvironment of a lysine residue, is the ideal source of information upon which a prediction can be made. Unlike past models in the literature that focused on features in protein sequences, the model architecture proposed by the present work is the first one that can exploit protein 3-d structure based features to a great extent.

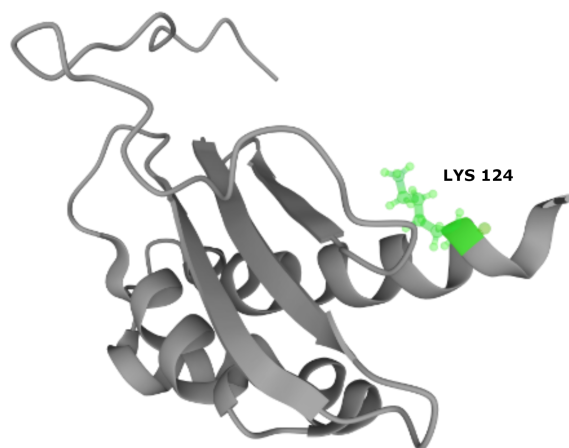


Fig. 1: 3-d structure of PDB 2KQK in the cartoon representation with lysine 124 highlighted.

2 RELATED WORK

THE application of computational methods such as Machine Learning or Deep Learning in predicting protein PTM sites is by no means novel. For simplification, a carbamylated site is called a positive and an uncarbamylated one is called a negative. G. Huang et al. 2013 proposed a Machine Learning model, one-class K Nearest Neighbour (KNN) for predicting carbamate sites with protein primary and secondary structure based features. The model was trained by 40 positive sites gathered from the Uniprot database with 70% sequence identity reduction to learn a discriminative boundary enclosing the positive class; any new sample that falls inside of the boundary in the feature space would be classified as a positive. A Leave-One-Out Cross Validation (LOOCV) was carried out, and the model obtained a 82.5% Sensitivity (Sn) and a 96.37% Specificity (Sp). On the hold-out test set, the model achieved a 66.67% Sn and a 100% Sp. However, the number of positives in this set was only 6.

Jimenez-Morales et al. 2014 presented the first model that used features extracted from protein 3-d structures. From the Protein Data Bank (PDB), they initially gathered 251 positive sites from proteins with at least one subunit containing a carboxylated lysine residue and 4259 negative sites which were the remaining lysine residues from the same proteins. Two sub-sets with redundancy reduction of the proteins was constructed using 40% and 90% sequence identity cut-offs. They designed the feature vector that was composed of occurrence numbers of amino acids, water molecules, and metal ions within a 5 Å Euclidean distance of a lysine

residue. Three Native Bayesian classifiers were trained and tested using the three datasets. They achieved high performance in the LOOCV with a 83.72% Sn and a 99.58% Sp using the set with 90% sequence identity reduction, performance using the other two sets was even higher. Yet, as reported by the authors, most of the carbamylated lysines they obtained were buried metal-stabilized ones. These sites are not biologically important in some cases, such as in the study of protein-CO₂ binding. Also, as suggested by Linthwaite and Cann 2021, models trained with mostly buried carbamates may only capture the characteristics of a subset of the possible carbamate sites.

Ning et al. 2021 presented another Machine Learning based method called pQLyCar that used protein sequence based features. They constructed a dataset which included 723 positive sites and 1197 negative sites with 60% sequence identity reduction from Uniprot, PLMD and dbPTM database. The idea behind pQLyCar is similar to the standard two-class KNN that when predicting for a query sample, K "nearest" samples will be selected to train a Support Vector Machine (SVM) classifier to classify the query one. The selection of the K samples is based on their Euclidean distances to the query sample in the feature space. They also obtained high performance in the LOOCV featured by a 96.49% Sn and 99.59% Sp, but no independent test set was used. While their dataset was much bigger in terms of the size of the positive class, a spreadsheet documenting the exact sites they used is unavailable. Therefore, the amount of buried lysines in the set is unclear.

A limitation of these past research is that they did not have a benchmark dataset to use. Hence, it is not possible to directly compare the performance of different predictors when they were trained and tested with drastically different data. Also, due to the lack of effective experimental methodologies, previous studies tended to use buried carbamylated lysines, which could potentially limit their models' generalizability. Moreover, far too little attention was paid to the data imbalance issue. In fact, none of the past works proposed a suitable strategy to directly deal with this problem. Furthermore, no previous study discussed the possible existence of false-positives or false-negatives in their datasets, and it may be owing to the fact that they mostly used the buried carbamate sites which were clearly observed at atomic resolution. Nevertheless, such consideration is necessary when the new data is applied. Finally, a potentially relevant discussion about high model performance can be viewed in 7.3.

While the existing predictors for identifying Carbamylation sites are all Machine Learning based, many Deep Learning models have been proposed in predicting other PTMs. For instance, to detect Succinylation sites, H. Wang et al. 2021 trained a neural network that applied both convolutional layers and Attention Mechanisms to make inferences based on protein sequences, physicochemical properties, and structures; Ahmed et al. 2021 attempted to use an LSTM Recurrent Neural Network (RNN) to process protein sequences to determine Phosphorylation sites. These Deep Learning methods are featured by their ability to learn features given certain inputs, instead of accepting features manually extracted by researchers. However, few of these studies have proposed a network architecture that is suitable

for protein 3-d structure based inputs.

The present work contributes to this area of research by:

- curating a benchmark dataset with new solvent accessible carbamate sites;
- exploring a neural network architecture that is ideal for 3-d structure based inputs;
- demonstrating the effectiveness of a method for dealing with data imbalance;
- discussing a possible solution to transform the uncertainty in the Ground Truth labels to additional knowledge and integrate it into the network training;
- presenting a trained neural network and its performance.

3 METHOD

3.1 Preparation of the Data

THE carbamate sites used in the present work come from three sources:

- unpublished experimental hits identified by the research group led by Prof. Martin Cann at Durham University;
- high confidence Lys-CO₂ sites from Supplementary Table 4 published by King et al. 2022;
- candidate Lys-CO₂ sites from Supplementary Data 1 published by King et al. 2022.

It should be pointed out that in the work of King et al. 2022, a statistical test (one-sample one-tailed student t test) was used to partially determine the confidence level of a candidate site. The high confidence ones were determined by various other criteria in addition to the outcome of the t test. To maximize the true-positive rate, the selection among the candidate sites was conducted only to the ones that obtained a p-value less than 0.05. The negatives were also gathered from the Supplementary Data 1 supplied by King et al. 2022. To refine the selection, same criteria was followed that only the lysines that obtained a p-value less than 0.05 from within a peptide chain that did not contain a candidate carbamate site were considered. In total 202 (with duplicates) carbamate and 1881 non-carbamate sites could be gathered from the sources.

The availability of the 3-d structures of these sites is critical since the structure based feature is one of the focuses of this study. All these proteins have structures predicted by a computational model—AlphaFold (Jumper et al. 2021) and stored in the AlphaFold Database. A number of them also have one or more structures archived in the Protein Data Bank (Berman et al. 2000). These structures were observed using experimental methods, such as X-ray crystallography, NMR spectroscopy, and electron microscopy. For the structures found in the PDB, alternate conformations (usually from NMR ensemble) and alternate side chain rotamers for each lysine were all considered useful. The reason for extracting and including multiple structural representatives is that proteins are flexible in terms of their structures, and including multiple structural arrangements could capture this flexibility. Here, a benchmark dataset can be constructed using these structures (see Table 1).

Data Imbalance—one remaining problem is how to deal with the non-negligible size gap between the data of the

positive and negative sites? Without a doubt, way more structures from the PDB and AlphaFold can be gathered for the negative class, resulting in an imbalanced dataset. As pointed out by Dou et al. 2021, data imbalance is a natural difficulty in PTM identification tasks that apply computational methods, and it usually results in the prediction bias towards the majority class. Methods for dealing with imbalanced data have long been proposed by the Machine Learning community (Chawla et al. 2002). In a data-level method, the minority class can be up-sampled using duplicates or synthetic samples generated by certain algorithms, such as SMOTE (Chawla et al. 2002), and the majority class can be down-sampled by random selection.

One preliminary solution is to down-sample the negative class, but not randomly. Specifically, for the negative class, only the structures predicted by AlphaFold were included into the benchmark set. Although the flexibility in the structures was not captured, the diversity of the negative class was still preserved. On the other hand, multiple structural arrangements for each positive lysine were all included into the set. In other words, in the benchmark set, the positive class is on the order of **n to 1** while the negative one is on the order of **1 to 1**. In total, the balanced benchmark set consists of 1376 positive samples and 1342 negative samples. The detailed composition of the set is displayed in Table 1. The training set were randomly selected from the benchmark set, containing **1107 positive and 1107 negative samples**. The remaining **269 positive and 235 negative samples** were treated as the test set. However, such a training set should be treated with caution due to the complexity in its composition, as will be discussed in 3.4.

3.2 Feature Extraction

3.2.1 3-d Structure based Feature

THE microenvironment of a lysine residue can be characterized by the amino acids surrounding it within a cut-off distance. This distance is the Euclidean distance between the terminal zeta (*NZ*) position of the target lysine and the alpha carbon (*C α*) of a surrounding amino acid. Empirically, the cut-off distance of 12 Å was selected though 20 Å was also considered. Inspired by the inputs of language models, such as a Transformer (Vaswani et al. 2017), a collection of amino acids can be mathematically described as a set of vectors with each vector representing one amino acid, the same way researchers represent a sequence of words in a sentence. The amino acid vector can be initially constructed using a common technique, One-Hot-Encoding. Specifically, Denoting ε_j as a vector with 1 at the *i*th entry and 0 elsewhere, given there exist 20 common types of amino acids, ε_j is a vector in $\mathbb{R}^{20}$. Each entry of ε_j corresponds to one amino acid type, and having a 1 at the *i*th entry suggests it is the *i*th amino acid. Here, the local environment of a lysine with *k* neighbouring amino acids within the cut-off distance can be represented by the set $\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_k\}$. Although the identity information has been encoded, the equally important positional information of amino acids in the environment was still missing. To accomplish this, a positional encoding method was used. This method has been empirically tested and proved effective in the domain of Scene Representation recently by Mildenhall et al. 2020. According to their implementation (Mildenhall et al. 2020), here γ is defined as a mapping from $\mathbb{R}$ to $\mathbb{R}^{2L}$, *p* is one

	Positive		Negative	
	Lysine	Structure	Lysine	Structure
unpublished experimental hits by Prof. Martin Cann’s group	37	451	3	3
supplementary table 4 (King et al. 2022)	24	82	-	-
supplementary data 1 [E.coli (hCit)] (King et al. 2022)	115	869	-	-
supplementary data 1 [Synecho (hCit)] (King et al. 2022)	26	44	-	-
supplementary data 1 [E.coli (all) & Synecho (all)] (King et al. 2022)	-	-	1878	1878
Total	202	1446	1881	1881
Total with reduction	161	1376	1342	1342

TABLE 1: The composition of the benchmark dataset. The set is composed of structures. For each carbamate lysine, it can have multiple structural representations included as samples in the set. The 3 unpublished negatives are resid 11, 27, and 29 of PDB 1UBQ. Among the positives, the reduction involves removing duplicated lysines across different sources; removing lysines that do not have a protein structure archived in the PDB and at the same time has a low confidence score in the structure predicted by AlphaFold. In total 161 distinct positive lysines and 1376 positive samples (structures) were left. Among the negatives, lysines that have a low confidence score in the structure predicted by AlphaFold were removed. In total 1342 distinct negative lysines and 1342 negative samples (structures) were left.

coordinate value, and L is a hyperparameter, formally the encoding formula is:

$$\gamma(p) = (\sin(2^0 \pi p), \cos(2^0 \pi p), \dots, \sin(2^{L-1} \pi p), \cos(2^{L-1} \pi p)). \quad (1)$$

It should be noted that the origin of the Cartesian space of a local environment was firstly set to the location of the target lysine. Then, the positional encoding function was applied to each of the three coordinate values. The parameter L was set to be 4. Accordingly, the positional information of an amino acid can be represented by a vector in $\mathbb{R}^{24}$. This vector was then concatenated with the 20-d one-hot vector. By now, the representation of an amino acid reflects both identity and relative position information. However, one may notice that the number of amino acid vectors can vary across different structures. In Deep Learning, the varying input shape is not an unsolvable problem. Recent development of the Self-Attention mechanism is the most widely applied solution as will be discussed (see 3.3).

3.2.2 pK_a and SASA

It is suggested that the pH of the local cellular environment plays a key role in regulating biological processes (Kilambi and Gray 2012). Predicting pK_a values to determine ionization states is important for understanding protein functions (Gokcan and Isayev 2022). Specifically, low pK_a values that allow for dissociation to a neutral amine and CO2 binding is the key to Carbamylation (Linthwaite and Cann 2021). Hence, the residue pK_a value was regarded as an important feature in predicting carbamate sites. The calculation of pK_a was based on a widely used python package called PROPKA (Olsson et al. 2011; Søndergaard et al. 2011). It should also be pointed out that a new Deep Learning based method for predicting pK_a (Gokcan and Isayev 2022) has been published shortly before the completion of the present work. Their method has been reported to have better performance than PROPKA in some ways and can be applied in future works.

SASA is in short for Solvent Accessible Surface Area. It is suggested that the amino acid side chain that can be modified by PTMs tends to be more accessible on the surface of a protein (Lu et al. 2011), in other words, has a higher

SASA. In an empirical study (Vandermarliere and Martens 2013), it was found that 90% of the residues that have undergone the Phosphorylation modification are solvent accessible. Solvent accessibility has also been used as a feature in a discussed past work (G. Huang et al. 2013). Accordingly, besides pK_a , SASA was considered to be another manually-extracted feature. The value of SASA was computed by a well-developed python package called Biobox (Rudden et al. 2022) using the Shrake-Rupley algorithm.

However, for each structure, the corresponding pK_a and SASA are two scalars which are not compatible with a set of amino acid vectors. The exact way that pK_a and SASA involve in the training and decision-making will be discussed in 3.3.3.

3.3 Neural Network Architecture

INSPIRED by functioning of human brains, a Neural Network (NN) is a mathematical model with multiple layers of neurons linked together (Emmert-Streib et al. 2020). One difficulty in developing an NN is dealing with inputs that may have different shapes. In classical architectures such as a Multilayer Perceptron (MLP), the set of parameters (neuron connections) is predetermined and fixed during the training. When the input changes in shape, for instance from L -dimensional to $L + 1$ -dimensional, new connections (parameters) between the information in the added 1 dimension and the existing neurons in the first layer are required, which however is not possible.

Vaswani et al. 2017 published the paper titled “Attention is All you Need”, a work that has long been recognized as a milestone in language processing and computer vision (lately). In simplified words, the Self-Attention Mechanism takes a set of vectors as the input and outputs a set of vectors with the same cardinality. Global Self-Attention also takes a set of vectors as the input but only outputs one single vector. Two things should be noted here: (1) vectors in a set do not have ordering; (2) the cardinality of the set can change. These good properties perfectly match the characteristics of the 3-d structure based features and render the Self-Attention block the foundation architecture of the proposed neural network (see Figure 2).

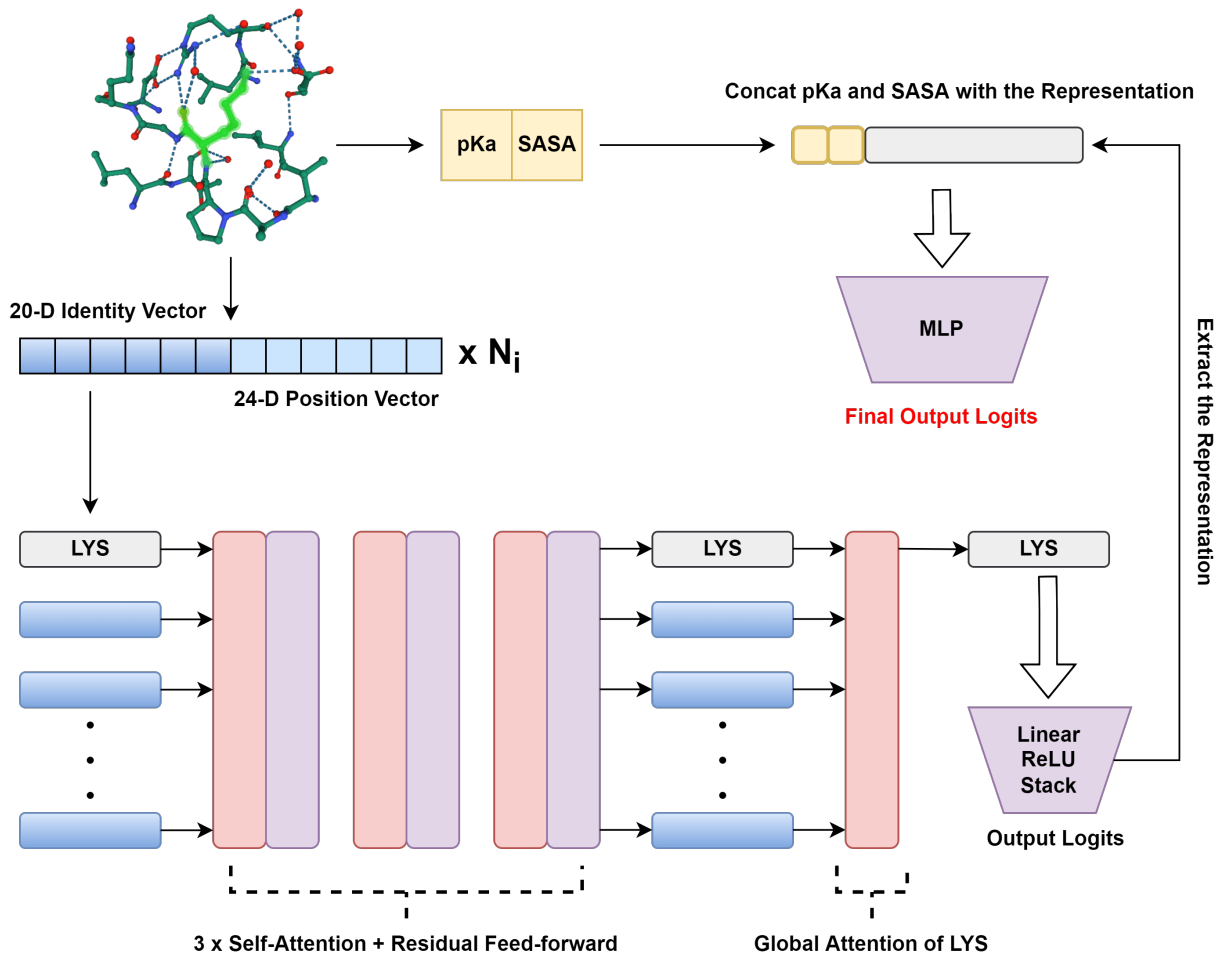


Fig. 2: The detailed input pipeline and the network architecture. For the local environment of a lysine residue i , in total N_i amino acid vectors are computed as the input of the network. Each vector's first 20 dimensions represent identity information, and the rest 24 dimensions reflect the relative position of the amino acid. Passing through 3 blocks of Attention and Residual Feed Forward, the N_i output vectors are then converted to one single vector by the Global Attention. The output of one hidden layer in the following Linear ReLU Stack is then extracted and concatenated with pK_a and SASA to form the input of an MLP which is responsible for generating the final output logits. The Attention Based NN and the MLP are trained separately.

3.3.1 Additive Self-Attention

ASSUMING there are N amino acids in the local structure of a lysine residue with each amino acid represented by a vector $\nu^j \in \mathbb{R}^h, j = 1, 2, \dots, N$, the outcome of one **Query** vector ν_q^j after Self-Attention is dependent on the N **Key** vectors which in this case is the whole set $\{\nu_k^1, \nu_k^2, \dots, \nu_k^N\}$. The output $\tilde{\nu}_q^j$ can be computed by:

$$\tilde{\nu}_q^j = \sum_{i=1}^N \text{softmax}(\alpha(\nu_q^j, \nu_k^i)) \nu_k^i, \quad (2)$$

in which $\alpha(\nu_q^j, \nu_k^i) = \nu_q^{jT} \tanh(W_k \nu_k^i + W_q \nu_q^j)$.

Intuitively, what Self-Attention does to a **Query** is firstly assessing the relationships (attention scores) between the **Query** and all the **Keys**, then integrating the information of the **Keys** based on these relationships (weighted summation) to the **Query**. Matrices W_k and W_q are trainable parameters, and the softmaxed values are the so-called attention scores of query-key pairs. When there is only one **Query** attending to the **Keys**, this operation is defined as the Global Attention. Naturally, an N by N (1 by N for

Global Attention) attention score matrix can be obtained by storing these attention scores in a matrix. What makes the attention matrix interesting is that it is unique for each microenvironment. An example of an attention matrix will be displayed (see Figure 7).

3.3.2 Residual Feed Forward

LEARNING abstract representations unusually requires a deep network architecture. Nevertheless, going deep does not always render a better result. He et al. 2016 first proposed a network framework called Deep Residual Net that has been universally applied in deep network architectures. The problem they attempted to address is that when building a really deep net, the optimization can be so difficult that the generalizability after extensive training can be worse than a shallow one. The solution He et al. 2016 presented is simply changing the original mapping $F(x)$ to $F(x) + x$ for some hidden layers. Intuitively, the optimization of the deep network in this case should be easier because if it gets hard, this so-called residual connection can make the deep net shallow. Specifically, the network

with residual connections can choose to push an actual mapping of a deep layer $F(x)$ to 0 and simply output x as if the deep layer does not exist. In the present network, the residual connections applied in the Residual block and the Self-Attention block can be described as:

$$\begin{aligned} \text{Residual Block: } F(x) &= \text{ReLU}(W_2(\text{ReLU}(W_1x)) + x), \\ \text{Self-Attention Block: } G(x) &= \text{layernorm}(\text{Attention}(x) + x). \end{aligned} \quad (3)$$

3.3.3 Integrating pK_a and SASA

TWO methods can be used to integrate the two scalars into the Self-Attention based net: (1) after the Global Attention, directly concatenate the two values to the output LYS vector; (2) train a new MLP whose input is the concatenation of a learned representation extracted from a layer of the final Linear ReLU Stack and the two values. Preliminary results demonstrated that when placing pK_a and SASA at the very end of the Attention based net using method (1), the final Linear ReLU Stack would converge quickly, leaving the Attention Blocks poorly trained. Hence, the second method is adopted as shown in Figure 2.

3.4 Weighted Random Sampling

TRAINING a neural network is just a simplified expression for updating the parameters of the net by stochastic gradient descent (SGD). The network gets updated every time after seeing just a small batch of data. The composition of every batch determines how well the net can be optimized. The conventional way of preparing these batches is by random shuffling and splitting of the dataset. However, such method can be problematic due to the complexity of the training data. The problem is that some positive lysines have contributed too many samples (in the extreme case, more than 200 for 1 lysine) in the training set while others have contributed too few (sometimes just 1). One bad possible example is that after the shuffling, several successive batches contains too many samples that belong to just one or two positive lysines. In this case, it is inevitable that the net will learn too much from these lysines, which may drastically decrease its generalizability.

To cope with this issue, a technique called weighted random sampling can be helpful. According to Efraimidis and Spirakis 2006, the weighted random sampling algorithm can be viewed as selecting m random items out of a population of size n with the probability of each item being selected determined by the item weight. In the context of the current task, the goal is to find a set of weights Ω such that when sampling 1 sample out of a population of size $N^+ + N^-$ (with replacement), in which N^+ and N^- represent the class sizes of the positives and the negatives in the training set, two conditions should be satisfied. The first condition addresses the equal likelihood of selecting a positive or a negative, let Θ^+ and Θ^- represent the positive and negative classes:

$$(a) P(x \in \Theta^+ | \Omega) = P(x \in \Theta^- | \Omega). \quad (4)$$

The second condition ensures the fairness when sampling a positive sample in a batch. Assuming there are k distinct lysines in the positive class and each lysine has its own set

Lys_i such that $\sum_{i=1}^k |Lys_i| = N^+$, for $i \neq j$ and $i, j \in \{1, 2, \dots, k\}$:

$$(b) P(x \in Lys_i | x \in \Theta^+, \Omega) = P(x \in Lys_j | x \in \Theta^+, \Omega). \quad (5)$$

It can be shown that the following construction of Ω can satisfy both conditions. For $i = 1, 2, \dots, k$, denote $n_i = |Lys_i|$, for $j = 1, 2, \dots, (N^+ + N^-)$:

$$w_j = \begin{cases} 1, & \text{when negative,} \\ \frac{N^-}{n_i k}, & \text{when positive and belongs to } Lys_i. \end{cases} \quad (6)$$

Proof. Let w_i^+ and w_i^- represent weights assigned to a positive and a negative. $P(x)$ is the unweighted probability of any one sample being selected, and $P(x | x \in \Theta^+)$ is the conditional probability of any one positive sample being selected:

$$\begin{aligned} (a) P(x \in \Theta^+ | \Omega) &= \sum_{i=1}^k \sum_{m=1}^{n_i} w_i^+ P(x) \\ &= \sum_{i=1}^k \sum_{m=1}^{n_i} \frac{N^-}{n_i k} * \frac{1}{N^+ + N^-} \\ &= \sum_{i=1}^k \frac{N^-}{k(N^+ + N^-)} \\ &= \frac{N^-}{N^+ + N^-} \\ P(x \in \Theta^- | \Omega) &= \sum_{i=1}^{N^-} w_i^- P(x) \\ &= \sum_{i=1}^{N^-} 1 * \frac{1}{N^+ + N^-} \\ &= P(x \in \Theta^+ | \Omega) \end{aligned}$$

To show the conditional probability is independent of i , for $i = 1, 2, \dots, k$:

$$\begin{aligned} (b) P(x \in Lys_i | x \in \Theta^+, \Omega) &= \sum_{m=1}^{n_i} w_i^+ P(x | x \in \Theta^+) \\ &= \sum_{m=1}^{n_i} \frac{N^-}{n_i k} * \frac{1}{N^+} \\ &= \frac{N^-}{kN^+} \quad \square \end{aligned}$$

3.5 Loss Design

THE Cross-Entropy (CE) Loss is widely used in binary classification tasks in Deep Learning. Denoting p as the predicted "probability" for the class with label $y = 1$, it can be described as:

$$CE(p, y) = \begin{cases} -\log(p), & \text{if } y = 1, \\ -\log(1 - p), & \text{otherwise.} \end{cases} \quad (7)$$

However, even when p is significantly greater than 0.5 and the prediction is correct, in other words, the prediction is easy, the loss $-\log(p)$ can still have "non-trivial magnitude" (Lin et al. 2017). In the present work, it is assumed that the negatives should be easy to predict after some amount of training because the negative set is more diversified and contains more information to learn. On the other hand,

the positives are hard to spot due to the lack of diversity. Another possibility is that some samples can share similar patterns which make them collectively relatively easier to predict. The implication is that when the losses generated by these easy predictions accumulate, the magnitude is non-negligible, which can downgrade the efficiency in optimization. Hence, it is necessary to answer the question—how to weaken the losses of the easy predictions but amplify the losses of the hard ones? Focal Loss introduced by Lin et al. 2017 can be the answer. Focal Loss can be described as:

$$FL(p, y) = \begin{cases} -\alpha(1-p)^\gamma \log(p), & \text{if } y = 1, \\ -(1-\alpha)p^\gamma \log(1-p), & \text{otherwise.} \end{cases} \quad (8)$$

Take $y = 1$ as an example, given γ is greater than 1, the term $(1-p)^\gamma$ adjusts the loss such that it shrinks drastically when p is already high (close to 1). α is another hyper-parameter for weighting the loss by classes. The two hyper-parameters γ and α are set to be 2 and 0.75, respectively.

3.6 Label Smoothing

THE final issue to address lies in the uncertain nature of the Ground Truth labels. Even though the (non)modification sites were gathered with great care, false-negatives and false-positives are impossible to rule out. The critical question to ask is how well can a neural network perform when its training involves certain amount of inaccurate labels?

3.6.1 The Student-Teacher Analogy

THE relationship between a neural network and the Ground Truth can be characterized by the student-teacher analogy. Naturally, the net is the student who learns from the teacher, the Ground Truth. In an experiment, Guan et al. 2018 intentionally corrupted 80% of the labels in the MNIST training set (a standard in handwriting recognition) and trained a neural network using the corrupted data to do the classification. The test performance using the correct labels was incredible that the error rate was only 8.23%. Guan et al. 2018 even titled their experiment by “Beating the Teacher”. They further pointed out the performance of the network would collapse when the amount of the corrupted labels increased to a certain percentage. The implication is that they empirically proved that the performance of a neural network is actually not bounded by how good the teacher is under certain condition. The collapse point is closely linked to the mutual information between the corrupted labels and the correct ones (Guan et al. 2018). While in the current task, the perfectly correct labels are unavailable, and therefore mutual information cannot be evaluated. However, the network performance as will be shown may suggest that the training data is far beyond the collapse point.

3.6.2 Soft Labels

AS initially proposed by Szegedy et al. 2016, label smoothing is a regularization method and has been shown to be effective for improving the performance of networks in image classification and language processing tasks (Müller, Kornblith, and Hinton 2019). The relevance of this technique to the current task is that intuitively, all it does

is adding a small amount of uncertainty to the Ground Truth labels. For instance, when computing the Cross-Entropy loss, instead of using the target label $[1, 0]$, a so-called soft label $[1-\varepsilon, \varepsilon]$ now is used, in which ε is a small number. This setting captures the real uncertainty existing in the Ground Truth and transfers it to extra information for the network to learn. Technically, according to the Müller, Kornblith, and Hinton 2019, training a neural network with soft labels forces the difference between the logit of the correct class and the logit of the incorrect class to be a constant. In other words, the network trained by soft labels is more robust and can not easily be over-confident (over-fitted) by pushing one logit too large and the other one too low. In the present network, the small number ε was set to be 0.03. As will be shown, with such a relatively small level of smoothing, the generalizability of the net can be improved.

4 RESULTS AND DISCUSSION

4.1 Contour Plot of SVM Using pK_a and SASA

USING pK_a and SASA as two features, a Support Vector Machine (SVM) with a Gaussian kernel was trained and regarded as the baseline model. The prediction values (probability) of the SVM for a grid of (x, y) coordinates were used to generate the contour plot as shown in Figure 3. One key observation is that as expected, a lower pK_a is associated with a higher probability of being a positive. At the bottom-left region of the scatters, the positives are more densely distributed than the negatives. Nevertheless, a residue being more solvent accessible does not actually render a higher probability of being modifiable since the positives and the negatives clearly overlap when SASA is high.

In general, the effectiveness of using an SVM to separate the two classes with pK_a and SASA is not exceptional and the large overlapping region (see Figure 3) may suggest a high false-positive rate in the real application. One final observation that should be noted is that there is a small cluster of negatives with pK_a values between 11 to 12. pK_a values for negatives were computed based on their structures predicted by AlphaFold. Such a cluster could potentially indicate a systematical error in the predicted structures and should be further investigated by researchers.

4.2 Effect of Weighted Random Sampling

TO visualize exactly what weighted random sampling did, 20 batches of data (batch size 64) were sampled using the two methods, random shuffling & splitting and weighted random sampling using the computed set of weights Ω . A comparison of data composition is depicted in Figure 4. With the conventional method, a few lysine residues contributed too many samples in the 20 batches. On the contrary, when weighted random sampling (with replacement) was applied, all the residues were equal likely to be sampled and the diversity of the residues in each batch should be improved.

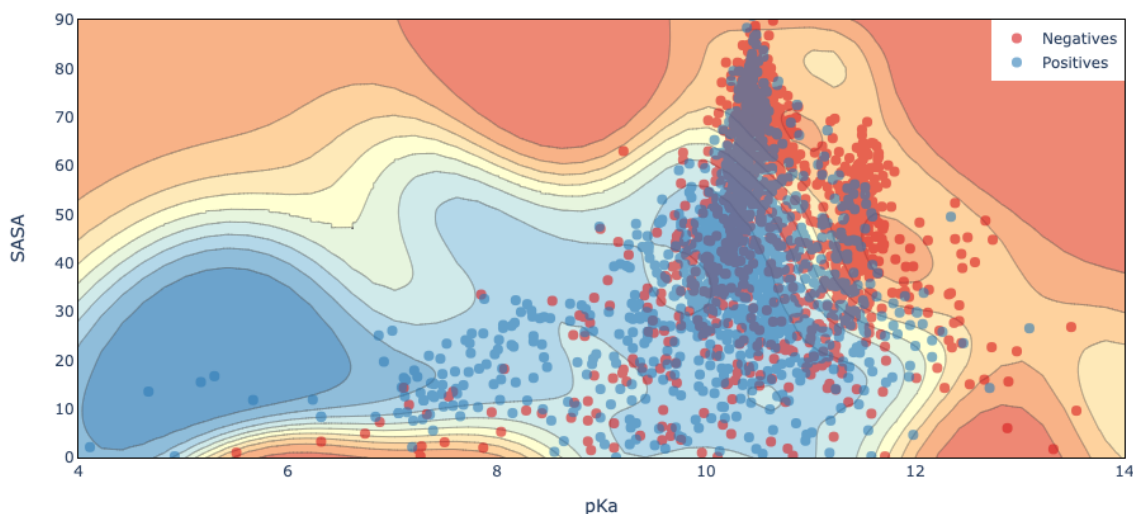


Fig. 3: Contour Plot of SVM Prediction Values and Scatter Plot of SASA against pK_a . Only the training data is used to generate the plot particularly to reflect how well the two classes can be separated by the Machine Learning algorithm and the two features. The blue region represents the area in which any points will be predicted as a positive. Points that fall into the red region will be predicted as negatives.

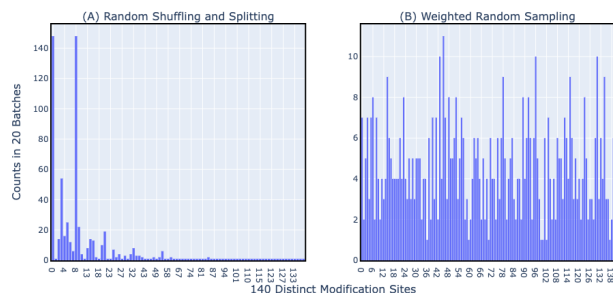


Fig. 4: The composition of 20 batches of data using two sampling methods. The x axis values are categories representing 140 distinct positive residues. The y axis indicates how many times each residue showed up in the sampled data.

4.3 Performance

INCLUDING the baseline model SVM, in total 6 models were trained and tested using the same training and test sets (see Table 2).

pK_a and SASA: the first observation is that SVM has already obtained high performance in terms of Sensitivity (76.6%). However, a low Specificity combined with a high false-positive rate suggest that a great number of negative samples overlapped with the positive samples in the pK_a -SASA feature space. An MCC of 0.44 indicates a mediocre correlation between predicted labels and true labels. Overall, it is evident that pK_a and SASA have certain predictive power, but additional features should be added to better separate the two classes.

Basic Network: the neural network trained with cross-entropy loss has obtained the worst Sensitivity. One possi-

ble explanation is over-fitting since no regularization techniques were used except for just a few dropout layers. However, the extremely high 76.2% Specificity suggests that the negatives were easier to classify. The net was likely to have learned more knowledge from negative sample but less from positive ones, which is not exactly the ideal outcome. However, it is reasonable because the negative class contains more distinct lysine residues, and naturally more knowledge to learn. Targeting the limitations, label smoothing and focal loss were subsequently implemented to improve the performance.

Label Smoothing: probably the most significant observation is that just a small amount of smoothing in the Ground Truth labels significantly improved the network's performance in all metrics by 3 to 4 percent. The overall accuracy jumped from only 71% to nearly 75% and the improved MCC of 0.5 suggests a stronger correlation between predictions and the Ground Truth. What can be inferred is (a) the slight amount of uncertainty injected into the labels was effectively learned by the net, *i.e.* the net learned more knowledge than the basic one; (b) the level of over-fitting decreased due to the application of label smoothing as a method of regularization. It can be concluded that label smoothing is an effective technique in dealing with the dataset.

Focal Loss: training a net with focal loss could improve Specificity substantially from 69.5% to 75.4%. It makes sense because spotting a positive originally can be hard and focal loss could force the net to learn harder from these difficult predictions. However, focal loss is not an ideal trick because the Specificity decreased from 80.1% to 73.2%. Nevertheless, the fact that applying focal loss can help researchers control the trade-off between Sn and Sp can be helpful depending

	Sn (%)	FP (%)	Sp (%)	FN (%)	Acc (%)	MCC
SVM with pK_a & SASA	76.6	27.0	67.7	28.4	72.4	0.44
NN CE	66.5	23.8	76.2	33.5	71.0	0.43
NN CE + LS	69.5	19.4	80.1	29.4	74.8	0.50
NN CE + LS + pK_a & SASA	79.9	25.6	68.5	25.1	74.6	0.49
NN FL + LS	75.4	23.7	73.2	27.7	74.4	0.49
NN FL + LS + pK_a & SASA	78.8	24.6	70.6	25.6	75.0	0.50

TABLE 2: Performance Comparison of Trained Models. Model acronym explanations: SVM—support vector machine; NN—Attention based neural network; CE—cross-entropy loss; LS—label smoothing; FL—focal loss. Performance metric explanations: Sn—sensitivity; FP—false-positive rate; Sp—specificity; FN—false-negative rate; Acc—total accuracy; MCC—Matthew’s correlation coefficient.

on the researchers’ needs.

Network with pK_a and SASA: it can be observed that when pK_a and SASA were added to the learned representations of the net with cross-entropy loss and soft labels, both the Sensitivity and the Specificity changed by around 10%. This drastic shift in the performance may suggest that the information contained in the learned representation was very different from the information carried by pK_a and SASA. Another key finding is by far, the test results of the MLP has officially outperformed the baseline model SVM in all metrics with a nearly 80% Sensitivity. On the other hand, the performance shift of the network with focal loss and soft labels after combining pK_a and SASA was more gentle, and the second MLP also obtained a better performance than the baseline model. Because the second MLP (last row in Table 2) earned higher scores in the total accuracy and MCC than the first one did (the third row in Table 2) did, it is regarded as the final optimal model (considering the Attention based net and MLP jointly as a model).

4.4 Embedding Visualization

REPRESENTATIONS extracted from the Attention based net can be viewed as low-dimensional embeddings of the network inputs. Specifically, the mapping can be described as $f: \mathbb{R}^{n \times k} \rightarrow \mathbb{R}^m$, in which n, k, m represent the number of amino acids, vector length, and embedding dimension, respectively. It is beneficial to visualize the learned embeddings in the 2-d space to truly understand how well the neural network abstracted the input information in order to separate the two classes. The embeddings were from the Attention based net with focal loss and soft labels. t-SNE (t-distributed Stochastic Neighbor Embedding), a commonly used dimensionality reduction algorithm in Machine Learning was applied to further learn a 2-d representation of the embeddings.

Training Set: as shown in Figure 5, the positives and negatives were relatively well-separated. The positives formed a curvy shape and were distributed at the bottom-left region while the negatives were more compactly distributed at the top-right region. The so-called minority positives are the ones associated with the positive lysine residues that only have few structure representatives (usually just 1) in the training set. The fact they were all separated from the negatives proves the net was not biased by learning too much from the positive residues with too many conformations; otherwise, these minority positives would

not be well-separated or could be falsely classified in the training set. Samples associated with the experimental hits supplied by the research group at Durham University were also marked by grey stars to give biologists a clearer picture.

Test Set: separability of the test set is less obvious as shown in Figure 6. It is undeniable that given the size of the dataset, even though regularization was applied, overfitting could still be an unsolvable issue. In general, the positives still formed a cluster, but more negatives were mixed into it. However, observations of these green stars are encouraging. They are the structure representatives of the positive lysine residues that do not have conformations included in the training set. 16 out of 22 so-called difficult positives were clearly separated and compactly distributed at the “right” place (bottom-left). Even though 6 of them were missed, the possibility of them being real false-positives may not be ruled out.

Although adding pK_a and SASA to the learned embeddings would bring more noise to the depicted representations, general separability of the training and test sets could preserve.

4.5 Attention Matrix Visualization

ATTENTION Matrices are collections of Attention Scores stored in the matrix form. Taking protein P0ACD4 (Uniprot) as an example, in its structure 2KQK (PDB), 28 amino acids can be identified in the microenvironment of lysine 124. The **Attention Matrix** (Figure 7) for this particular instance is 29 by 29 (the lysine itself is also included) with each entry being a measurement of the relationship between two amino acids. Just a variation of the Attention Matrix, the **Global Attention Matrix** is 1 by 29.

Attention Matrices can tell how the input amino acid vectors gather information from each other. For instance, in Figure 7, information of ASN and ALA (two dark green columns) were largely gathered by all the amino acids because they obtained higher attention scores with respect to all the amino acids. Knowing the flow of information allows researchers to examine the interpretability of the network decision-making for lysine residues of particular interest.

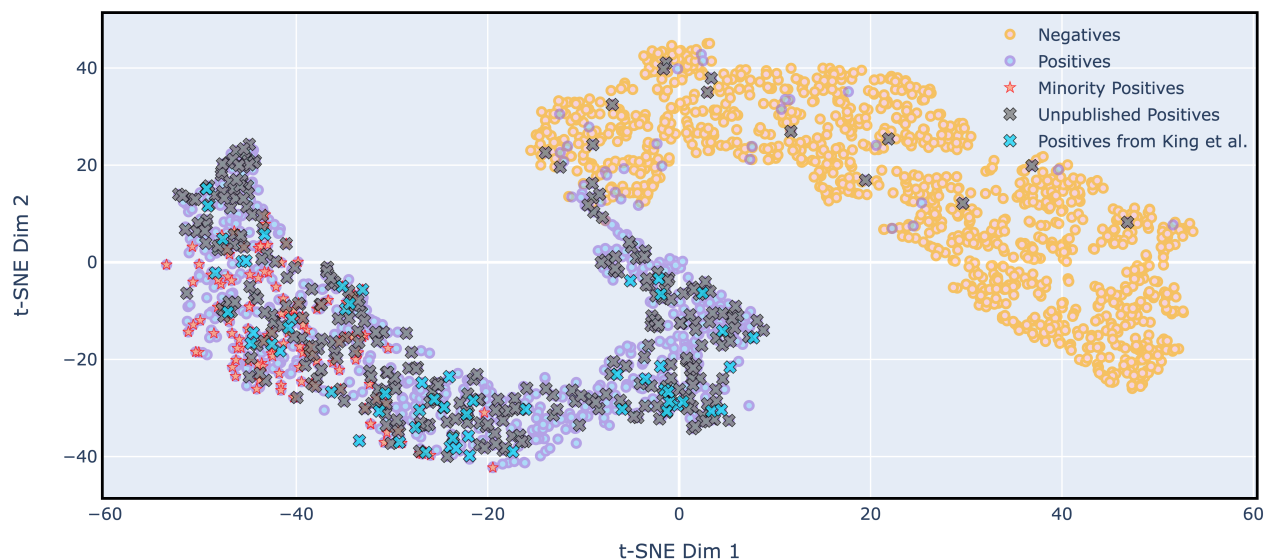


Fig. 5: Training set t-SNE visualization of embeddings learned by the Attention based net with focal loss and soft labels. Minority positives were samples of positive lysine residues that only have few conformations in the training set. Unpublished sites are samples of the carbamate sites identified by the research group at Durham University. Sites from King et al. 2022 are samples of the 24 high confidence Lys-CO₂ sites.

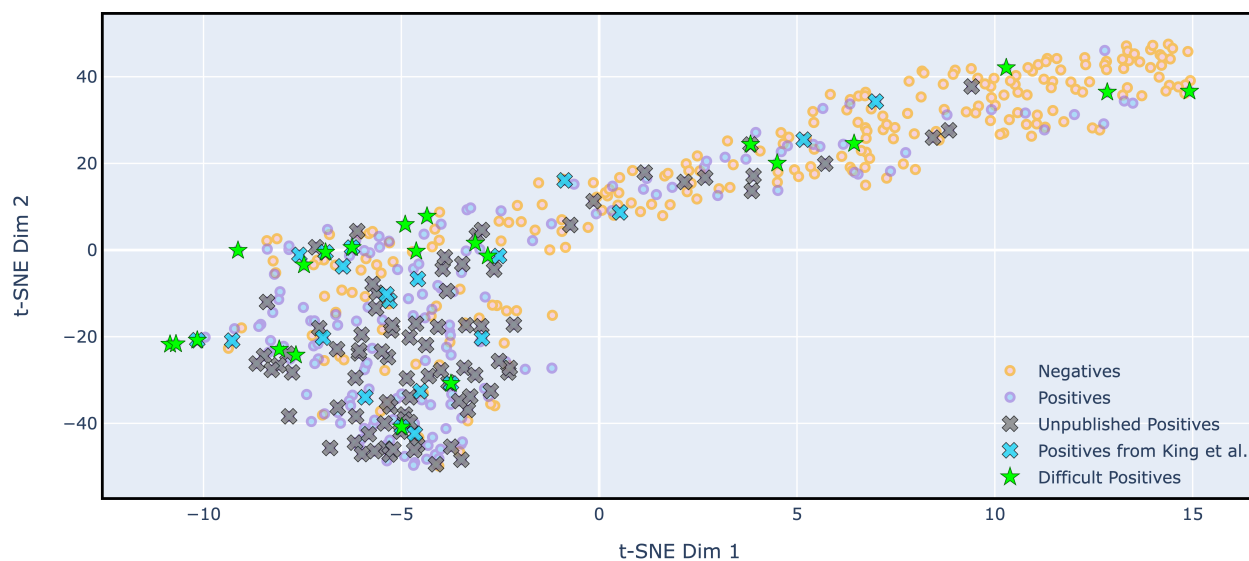


Fig. 6: Test set t-SNE visualization of embeddings learned by the Attention based neural network with focal loss and soft labels. Difficult positives are samples of positive lysine residues that do not have conformations in the training set. The 6 unidentified difficult positives were [PDB:1UBQ,63], [P0AFM6,PDB:4WHE,21] (duplicated), [P73452,predicted,384], [P08390,predicted,178], [P17846,predicted,519].

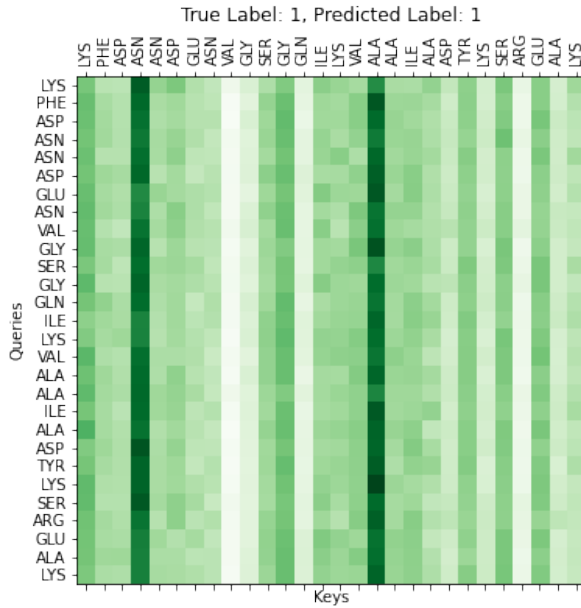


Fig. 7: Visualization of the First Attention Matrix for protein P0ACD4 (Uniprot), structure 2KQK (PDB), resid 124. Each entry of the matrix represents a measurement of the relationship between two amino acids. The darkness of the entry's color suggests how strong the relationship is. Each amino acid gathers information mainly from strongly-related amino acids.

4.6 Predicted Sites in Cyanobacteria and Human Proteins

CYANOBACTERIA are likely the first organisms that treated CO₂ as the primary source of carbon (King et al. 2022). Also, several hundred lysine residues was identified to have heightened reactivity in the human proteome by global profiling (Hacker et al. 2017, as cited by King et al. 2022). Following these findings, as the final step of the work, the trained Attention based net (the second-to-last row in Table 2) was applied to lysine residues in cyanobacteria and human proteins to evaluate the prevalence of Carbamylation. The Attention based net was solely used without the corresponding MLP due to efficiency consideration. All the proteins were reviewed ones identified from the Uniprot database. Due to the huge quantity of the existing lysine residues in these proteins, only structures predicted by AlphaFold were used to generate the network inputs. In total, **1169197** lysine residues from **43181** human proteins and **210466** lysine residues from **13698** cyanobacteria proteins were gathered. Preliminary results demonstrated that **29.75%** of the lysine residues in human proteins and **28.9%** of the residues in cyanobacteria proteins were predicted as potential carbamate sites (see Table 3).

	Num of proteins	Num of lysines	Predicted (%)
Human	43181	1169179	29.75
Cyanobacteria	13698	210466	28.9

TABLE 3: Predicted carbamate sits in human and cyanobacteria proteins. The percentage indicates the amount of predicted carbamate sites among all the lysines.

5 LIMITATIONS AND FUTURE WORKS

5.1 Alternative Positional Encoding Method

ONE key step of leveraging information in protein structures is to preserve the relative positional information of each amino acid in the environment. The present work has only tested one type of positional encoding used in Scene Representation tasks. Other research (Parmar et al. 2018; Carion et al. 2020) that attempted to apply a Transformer based network to image classification or object detection tasks have followed the original positional encoding method used by Vaswani et al. 2017 and extended it to the 2-d case. As done by Parmar et al. 2018, an easy way is that supposing the one-hot-encoded (identity) vector is L -dimensional, one can use $d = L/3$ of the dimensions to encode x , y , and z coordinate values, respectively and add (or concatenate) the position-encoded L -dimensional vector back to the original L -dimensional one-hot vector. Yet, the tricky part is developing a way to convert the continuous 3-d coordinates to a discrete coordinate system (the position of a pixel in an image or a word in a sequence is discretely represented). One way to achieve this is by putting the environment into a 3-d box made up of unit boxes. Then, the position of an amino acid can be instead encoded by the position of the unit box in which the amino acid stays using the following formulas, pos represents the position and i represents the dimension:

$$\begin{aligned} PE(pos, 2i) &= \sin(pos / 10000^{2i/d}), \\ PE(pos, 2i + 1) &= \cos(pos / 10000^{2i/d}). \end{aligned} \quad (9)$$

5.2 Transfer Learning

BIG data is critical to Deep Learning. However, in the present task, it is impossible to obtain thousands or more carbamate sites. One idea is to use transfer learning to leverage the possible big data of other PTMs. According to Weiss, Khoshgoftaar, and D. Wang 2016, transfer learning is the process of improving the model's performance in the target domain with the help of the information learned in the source domain. Firstly, one can train the same Attention based net using the data of other non-enzymatic lysine PTM sites, such as Glycation. Secondly, one needs to freeze the parameters in the Attention and Residual Feed Forward blocks and further train (fine-tune) the remaining parameters (mostly just Linear ReLU Stack) using the Carbamylation data. Depending on the performance, one can further unfreeze other parameters. In this way, the learned knowledge (mostly likely the network's perception of a 3-d environment) from the Glycation data can be used to enrich the knowledge that can be learned from the Carbamylation data and improve the performance. It should also be pointed out that using the network architecture to solely study the Glycation PTM can also be beneficial.

5.3 Hyperparameter Optimization

HYPERPARAMETERS are model configurations that can be tuned to improve model performance. Due to time and computing power constraints, no grid search of the optimal collection of hyperparameters was implemented in the present study. Therefore, the values of these parameters

can possibly be further optimized.

Network Depth—three times Attention and Residual Feed Forward blocks were applied by the present work. Shallower networks with one or two times of these blocks were also experimented, but yielded poor test performance. Building a deeper network to learn more abstract representations is only advisable when more data is available.

Shapes of W_k and W_q —the W_k and W_q shown in equation 2 can take different sizes as long as the two matrices share the same size. The size applied by the present work is 24×44 , in which 44 is the length of an encoded amino acid vector. It should be pointed out that increasing the number of rows (e.g., 44×44 or even 96×44) would result in poor model performance.

Hidden Layer Size in Residual Feedforward Blocks—in the Residual Feedforward block (equation 3), an amino acid vector (output of Self-Attention) will go through two hidden layers: the first layer W_1 can change the vector size; the second layer W_2 will restore the vector size back to 44. In practice, the number of neurons in the first layer (i.e. number of rows in W_1) was 24, the same value as the number of rows of W_k and W_q discussed above.

Level of Smoothing—the value ε was set to be 0.03 in the present study; $\varepsilon = 0.1$ is also applicable but may require more epochs of training to yield good performance.

Focal Loss Parameters—the two parameters γ and α were set to be 2 and 0.75, respectively; tuning the value of α could impact the balance between Sensitivity and Specificity; the value of γ is not encouraged to change.

5.4 Alternative Way of Defining Adjacency of Amino Acids and Cut-off Distance

ADJACENCY of an amino acid with respect to a lysine residue in its microenvironment is defined by having a Euclidean distance of less than 12 Å between the NZ position of the lysine and the alpha carbon ($C\alpha$) of the amino acid. Nevertheless, it could be the case that for an amino acid, its $C\alpha$ is outside of the cut-off region while its side chain falls inside. The amino acid will be excluded in that case, which may not be the best choice. Therefore, an alternative way—including an amino acid if any of its atoms is within the cut-off distance can be further tested. Due to the time constraint, preliminary investigations were carried out as shown in Appendix (7.2).

6 CONCLUSION

CARBAMYLATION is a critical lysine-based Post-translational Modification (PTM). However, in proteins, the lysine residues that can be modified are hard to identify using conventional experimental methods. The present work has explored the possibility of using a Deep Learning model to accomplish the identification task with the help of a number of newly-identified carbamate sites.

This project is original in the following ways. Firstly, this work has filled a gap in the literature by presenting the a curated benchmark dataset with newly identified solvent accessible modification sites. Secondly, the present work has introduced a way to exploited protein 3-d structure based information by designing the input feature that gathers information of neighbouring amino acids in a lysine microenvironment. Two biologically important values pK_a and SASA have also been studied and incorporated into the model-training and decision-making by the proposed way. The Self-Attention based network architecture has been implemented to accommodate the input features with varying shapes. Thirdly, methods to effectively deal with data imbalance and potential noise in the Ground Truth labels have been discussed. Finally, in total 6 models, including one baseline Machine Learning model and five Deep Learning models were trained and tested. The final optimal model obtained a 78.8% Sensitivity, a 70.6% Specificity, and a 75% total accuracy. Large-scale computations applying one of the trained model have predicted that nearly 30% of the lysine residues in human and cyanobacteria proteins are modifiable. Future works can experiment with an alternative positional encoding method for better encoding amino acid relative position information. The idea of transfer learning can also be tested in order to cope with data deficiency. The Attention based network architecture can also be applied to study other non-enzymatic lysine PTMs, such as Glycation. An alternative way to define amino acid adjacency in determining neighbouring amino acids in lysine microenvironments can be further implemented.

DATA, MODEL, AND CODE AVAILABILITY

The code implementation of the present study can be found at GitHub: <https://github.com/manhao9843/AttentionBasedNetwork>. Trained models and data are available from the Degiacomi and Cann groups upon request.

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7 APPENDIX

7.1 Network Training Details

Batch Size—the applied batch size is 64. Smaller batch sizes including 32 and 16 were also tested by yielded poor performance.

Optimizer—Adam in the default settings was chosen; SGD with a momentum was also tested but was outperformed by Adam regarding the speed of convergence and test performance.

Learning Rate—In the first 40-60 epochs, the learning rate lr upper bound of Adam was the default value $1e-3$. After at most 60 epochs, the lr upper bound was adjusted to $1e-4$ to train 20-30 more epochs.

Number of Epochs—the ideal total number of epochs is 60 (more epochs may increase the rise of over-fitting). However, further training using a small learning rate such as $1e-4$ is encouraged if the test performance is relatively poor only when label smoothing is applied. The Attention based net with focal loss and label smoothing was trained by 60 epochs using $lr=1e-3$ and subsequent 20 epochs using $lr=1e-4$.

pK_a and SASA—the present work trained the Attention based net first without incorporating pK_a and SASA. In a preliminary experiment, pK_a and SASA were concatenated to the output of Global Attention and were involved in the training of the Attention based net. Embedding visualization revealed that the two classes were poorly separated on the training set. One possible explanation is that this network setting caused the final Linear ReLU Stack to converge quickly, leaving other core blocks untrained. Therefore, it is advised to incorporate pK_a and SASA only to the training of MLP.

7.2 Amino Acid Counts for Analyzing Microenvironments

ONE way to explore lysine microenvironments is by counting occurrences of amino acids within the 12 Å cut-off distance. Because these counts are highly similar across conformations of each unique lysine residue, to avoid duplicates, only structures predicted by AlphaFold in the training set were used. For each 3-d structure, 20 occurrence numbers that correspond to 20 types of amino acids can be obtained. Treating the occurrence numbers as new features, distributions of each of the 20 features conditioned by class labels (*i.e.* positive and negative classes) can be plotted by histograms. Two figures were produced, one applying the original definition of amino acid adjacency (Figure 10), one featuring the alternate definition (Figure 11).

In Figure 10, differences between conditional distributions are subtle. It may be owing to the fact that with the original definition of adjacency, amino acids whose side chains are in the microenvironment but alpha carbons are far away were not counted. However, Figure 11 with the alternate definition by which more relevant amino acids were included still shows no significant difference between

conditional distributions for all 20 types of amino acids. Nevertheless, with the alternate way, more amino acids, such as negatively charged ASP and GLU were observed in the microenvironments. Other residues including ALA, ILE, LEU, and VAL were also observed more frequently.

7.3 Good Performance or Good Hacking

HACKING refers to training a model that can obtain exceptional test performance but may perform poorly in real applications. Here (Table 4), three Machine Learning models were trained using the entire training set with the 22-d features— pK_a , SASA, and the 20 amino acid counts discussed in 7.2.

All three model obtained extremely high performance on the test set. The best one, the Random Forest even scored a 97.4% Specificity and a nearly 90% Sensitivity. The lowest 2.4% false-positive rate further makes it potentially the ultimate solution for Carbamylation. But why is this hacking?

	Sn (%)	FP (%)	Sp (%)	FN (%)	Acc (%)
Native Bayesian	83.6	20.2	75.7	19.8	80.0
Support Vector Machine	82.9	8.6	91.0	17.7	86.7
Random Forest	89.6	2.4	97.4	10.1	93.3

TABLE 4: Performance comparison of three Machine Learning models trained using the entire training set. The features were the 22-d vectors composed of pK_a , SASA, and amino acid counts. The Random Forest was based on 200 individual trees. The SVM applied a Gaussian kernel with the level of penalty $C=100$. The Native Bayes was a Gaussian one.

The key lies in how the training and test data were constructed and what features were used. The benchmark dataset was essentially composed of protein conformations, and a small portion of these conformations were randomly selected as the test set. The feature vector was composed of mainly amino acid counts that were highly similar across conformations of the same lysine residue. The consequence of this setting is that in the test set, a great number of samples became extremely easy to predict because their alternate conformations were seen by the model in the training set. Therefore, only the difficult positives introduced in 4.4 that were by no means involved in the training set can be used to really evaluate the performance of the three models (see Figure 8). It turns out that only 3 out of 22 of these difficult positives were identified, which apparently contradicts with the seemingly high model performance. The separability of the training set is also poor (Figure 9). Therefore, the three models that did not properly address this issue are not applicable and are the results of good hacking. It should be pointed out that the present work is not completely immune to the issue, however great care has been taken. The design of the input feature largely ensures the variability across conformations since not only amino acid identities but also relatively positives that can drastically vary across conformations were gathered. Also, the fact that 16 out of 22 difficult positives were spotted by the network is a strong indication of its true performance.

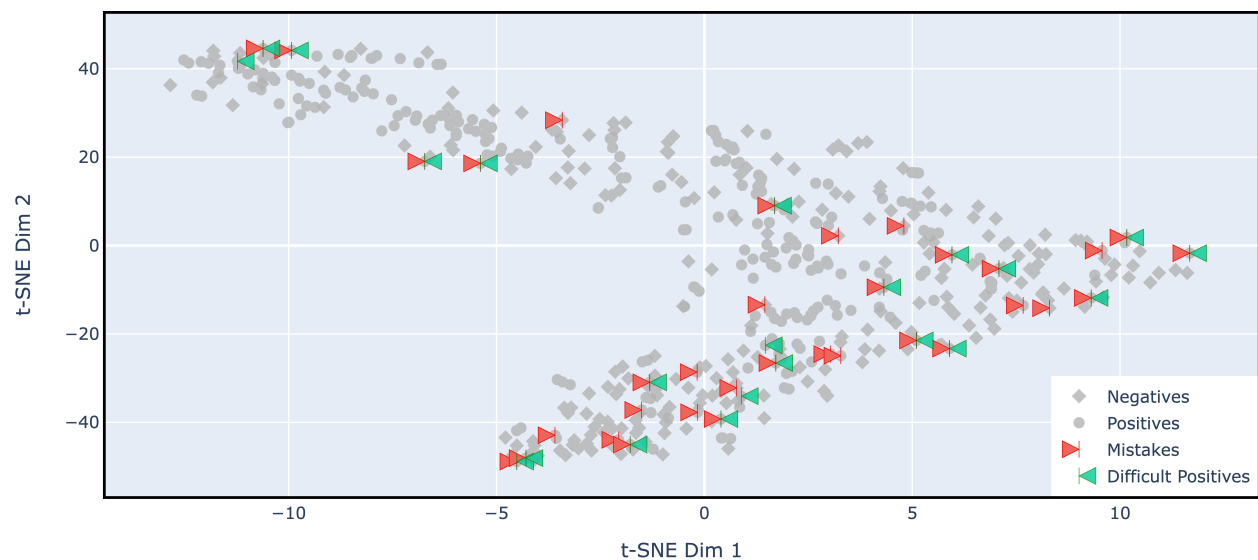


Fig. 8: Test set t-SNE Visualization with the 22 features. Prediction mistakes of Random Forest on the test set were marked by red right arrows. Difficult positives were marked by green left arrows.

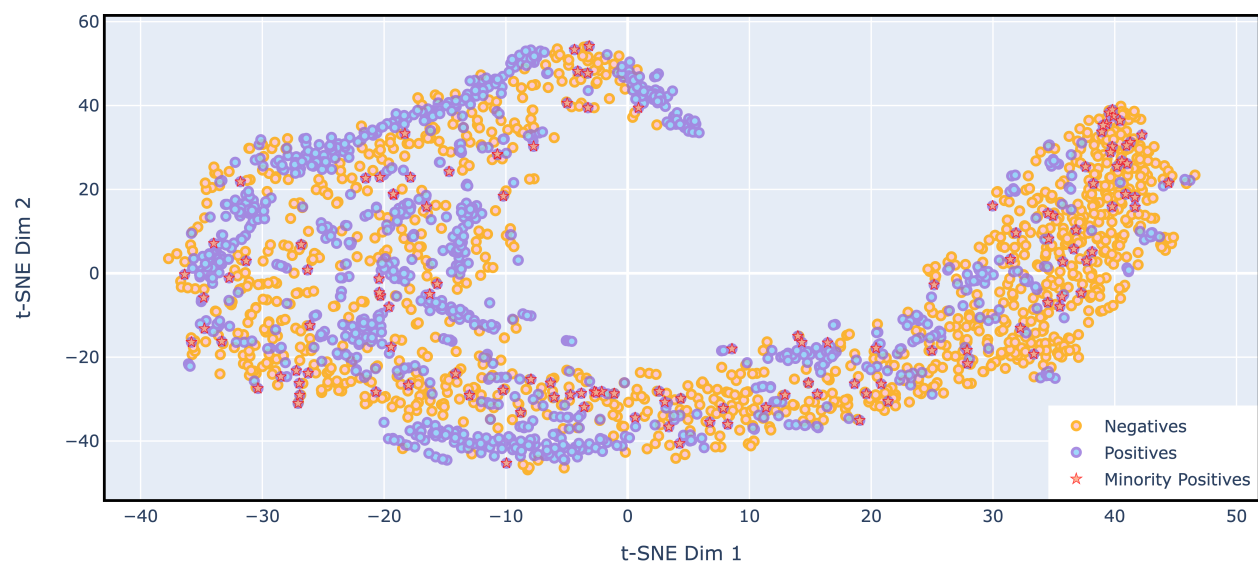


Fig. 9: Training set t-SNE Visualization with the 22 features. Minority positives marked by red stars were samples of lysines residues that only have few conformations in the training set.

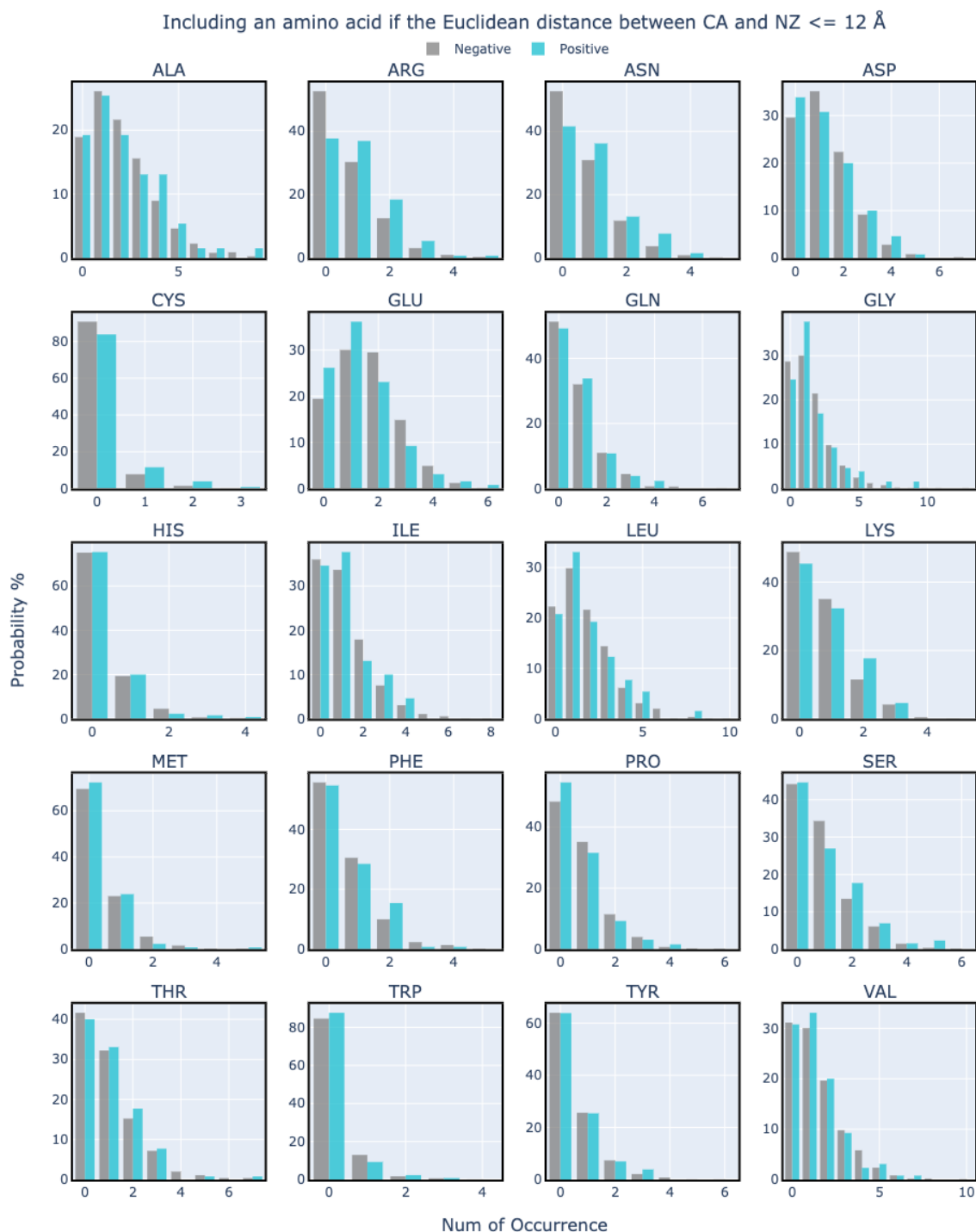


Fig. 10: Amino acid adjacency is defined as a less than $12 \text{ \AA}$ Euclidean distance between the $C\alpha$ of an amino acid and the lysine NZ position. Conditional Categorical Distributions of 20 types of amino acids. The x axis represents occurrence numbers for an amino acid in lysine microenvironment. The y axis is the normalized frequency in the percentage term (i.e. for one condition distribution, y values add up to 100).

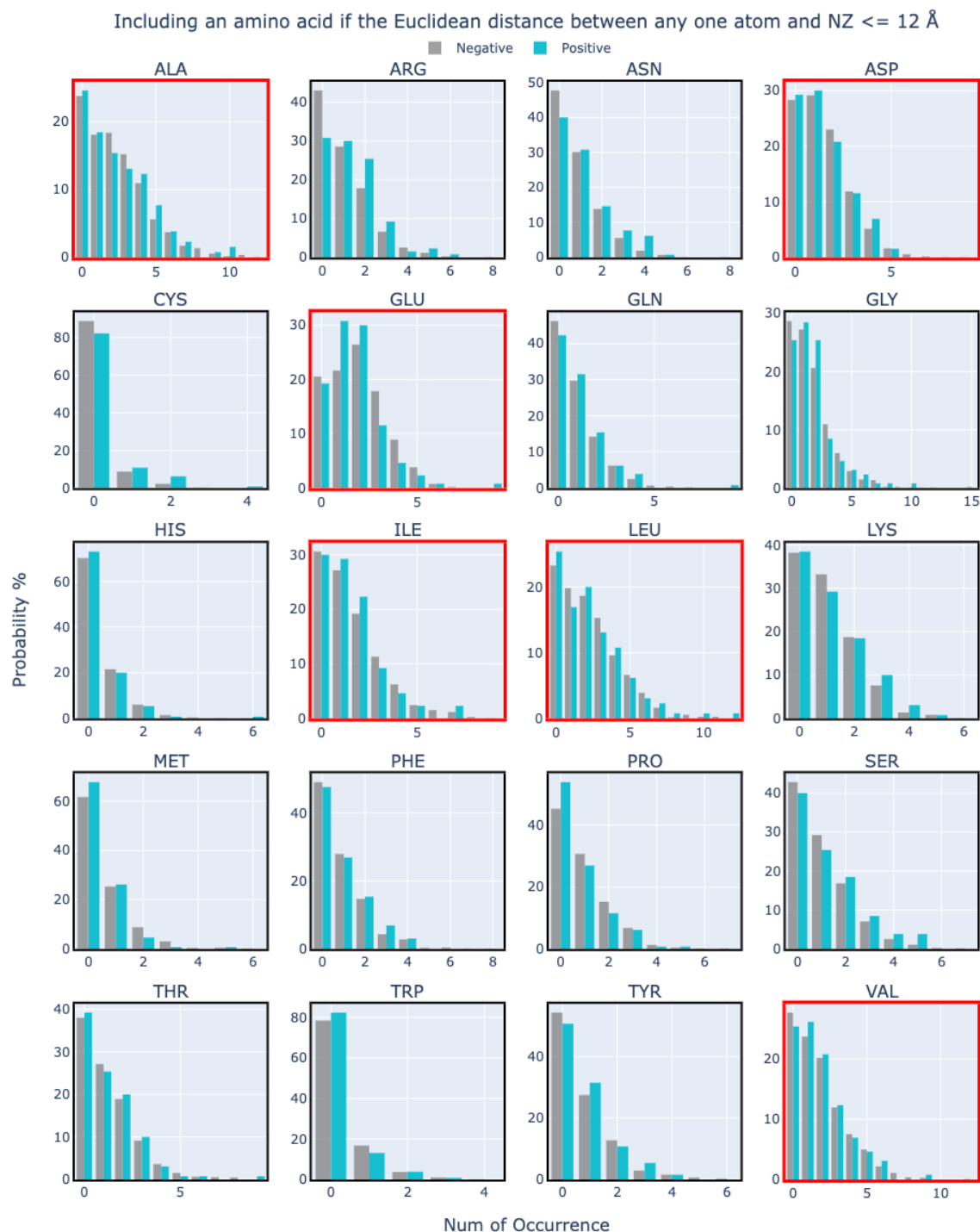


Fig. 11: Amino acid adjacency is alternatively defined as a less than $12 \text{ \AA}$ Euclidean distance between as least one atom of an amino acid and the lysine NZ position. Conditional Categorical Distributions of 20 types of amino acids. The x axis represents occurrence numbers for an amino acid in lysine microenvironment. The y axis is the normalized frequency in the percentage term (i.e. for one condition distribution, y values add up to 100). In comparison to Figure 10, amino acids that were more frequently observed were highlighted by red lines.